12 Inference - Bootstrap Uncertainty, Multiplicity Control, and Analysis Status

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0.1 Introduction

Every result reported so far is a point estimate. This notebook attaches uncertainty to those estimates and controls the error rate incurred by testing many of them at once. Three things are produced:

1. **Cluster-bootstrap confidence intervals** for every headline and subgroup estimate. Person-hours are nested within ICU stays, so the resampling unit is the stay: naive row-level intervals would be far too narrow.

2. **Multiplicity control.** The six pre-registered hypotheses form a *confirmatory* family and are corrected by Holm–Bonferroni at $\alpha = 0.05$ (FWER). The subgroup analyses and the alternative-label comparison form *exploratory* families (E1 subgroup discrimination, E2 subgroup label sensitivity, E3 alternative-label contrasts) and are corrected by Benjamini–Hochberg at $q = 0.05$ (FDR). The two procedures are deliberately different: a confirmatory claim must survive strict family-wise control, whereas an exploratory scan is intended to generate hypotheses and only needs its false-discovery proportion bounded.

Table 1: Internal temporal test performance with 95% stay-level cluster bootstrap intervals (Variant B)

model	n_events	auroc	auroc_ci	auprc	auprc_ci
primary	1040	0.7590	[0.743, 0.775]	0.00966	[0.0082, 0.0123]
gbt	1040	0.7424	[0.727, 0.758]	0.00877	[0.0075, 0.0110]
news2	1040	0.6415	[0.625, 0.659]	0.00325	[0.0029, 0.0037]
cox	1040	0.6444	[0.627, 0.661]	0.00333	[0.0030, 0.0038]
qsofa	1040	0.6091	[0.593, 0.625]	0.00296	[0.0026, 0.0033]
sirs	1040	0.6114	[0.595, 0.628]	0.00327	[0.0028, 0.0039]

3. **An analysis register** tagging every reported analysis as confirmatory, exploratory, or descriptive, with the pre-registration clause that licenses it. The register is written to disk so the thesis tables can carry the tag in the table itself rather than relying on a blanket methodological statement.

Status of this notebook. The uncertainty and multiplicity plan is *Protocol Amendment 1* (`00_preregistration.md` §13), written after the analyses in notebooks 02–11 had been run. The amendment is therefore itself post-hoc. It does not change any hypothesis, threshold, or estimand; it only attaches uncertainty to contrasts that were pre-specified and imposes a correction that the original protocol omitted. Where a correction changes a verdict, that is reported as the corrected verdict.

Family E3 belongs to *Protocol Amendment 2* (`00_preregistration.md` §§16–17), which adds two non-Sepsis-3 label definitions. It too is post-hoc, and it likewise leaves every pre-registered hypothesis untouched: variants D and E are never pooled with A–C, so H1 and the variance decomposition are computed from exactly the same inputs as before.

0.2 Data Assembly

The temporal test predictions for each label variant are joined to the comparator predictions and to stay-level demographics. Rows are ordered by `stay_id` so that a stay maps to a contiguous block, and each stay is given an index into a **common resampling frame** shared by variants A, B and C. Because all three variants are built on the same 12,132 test stays, one replicate draw can be applied to all three simultaneously, which makes the label-variant contrast in H1 a *paired* comparison.

```
## Common test-stay resampling frame: 12132 stays

##   Variant A: 548,854 person-hours, 1015 sepsis onsets
##   Variant B: 548,827 person-hours, 1040 sepsis onsets
##   Variant C: 504,828 person-hours, 2128 sepsis onsets
```

0.3 Internal Performance with Confidence Intervals

AUROC and AUPRC for every model on the Variant B temporal test set, with percentile intervals from 2,000 stay-level bootstrap replicates. A single replicate is shared across all six models, so the intervals are mutually consistent and model-to-model differences can be read off the same replicates as paired contrasts.

0.4 Confirmatory Family: the Six Pre-Registered Hypotheses

Each hypothesis is expressed as a single scalar contrast θ tested against its **pre-registered** threshold θ_0. Because the protocol states each hypothesis directionally, the tests are one-sided. P-values are obtained by recentring the bootstrap distribution on the null boundary and reading off the tail area, with the $(1 + \#)/(B + 1)$ convention so that no p-value is reported as exactly zero.

0.4.1 Contrast Estimation

H1 and H6 are drawn from the same replicates. Both compare quantities measured on one set of test stays, so a single resampling of stays serves both, and the resulting intervals keep the correlation between the spreads being differenced rather than treating them as independent.

```
## H4: random AUROC = 0.7424, temporal AUROC = 0.7590, delta = -0.0165
```

```
## H5: internal AUROC = 0.7590, external AUROC = 0.6960 [0.605, 0.778], degradation = 0.0630
```

```
## H2 instability by feature kind:
##              kind n_terms n_sign n_mag n_unstable pct_unstable
##            <char>   <int>  <int> <int>      <int>        <num>
## 1:  physiological      14      1     1          2         14.3
## 2: action-derived       8      1     4          4         50.0
## H2: of 22 shared covariates, 2 reverse sign, 5 change magnitude >=2x, 6 do either
## H2 uncertainty diagnostic: of the 6 flagged covariates, 3 exceed their pooled cluster-robust
##                   term max_diff pooled_se     z exceeds_noise is_action
##                 <char>    <num>     <num> <num>        <lgcl>    <lgcl>
## 1:        wbc_measured   0.8020    0.0517 15.51          TRUE      TRUE
## 2:  platelets_measured   0.5610    0.0529 10.60          TRUE      TRUE
## 3: creatinine_measured   0.2534    0.1024  2.48          TRUE      TRUE
## 4:           platelets   0.0002    0.0002  1.07         FALSE     FALSE
## 5:        vaso_anyTRUE   0.0645    0.0864  0.75         FALSE      TRUE
## 6:     bilirubin_total   0.0034    0.0056  0.61         FALSE     FALSE
```

```
## H3: 0 sepsis onsets in 179,003 pre-treatment person-hours; exact 95% upper bound on the per-h
```

```
## H3 (empirical counterpart, Variant F): 71 sepsis onsets in 178,303 pre-treatment person-hours
```

0.4.2 Corrected Confirmatory Verdicts

Each hypothesis becomes one row: the contrast, its interval, the null it is referred to, and the raw p-value Holm will correct. The two spreads are persisted as data here rather than left inside H1's free-text note, so the thesis table reads them instead of transcribing them by hand.

```
##    quantity estimate  ci_lo  ci_hi
##      <char>    <num>  <num>  <num>
## 1:    label   0.0123 0.0032 0.0316
## 2:    model   0.1499 0.1339 0.1707
```

0.5 Exploratory Family E1: Subgroup Discrimination

Subgroup AUROC is estimated with a bootstrap **stratified by subgroup level**: stays are resampled within each level, so the level sizes are held at their observed values. This is the correct design when the subgroups themselves are the estimand rather than a random draw.

Table 2: Confirmatory family: six pre-registered hypotheses, one-sided cluster-bootstrap tests against their pre-registered thresholds, Holm-Bonferroni corrected at family size 6

H	claim	estimate	CI	H0
H1	Label dominance	-0.1375	[-0.162, -0.108]	theta <= 0
H2	Coefficient instability	6.0000	–	not testable (threshold rule on point estimates, not a co
H3	Treatment leakage	0.0000	–	not testable (outcome absent from the window)
H4	Split optimism	-0.0165	[-0.037, 0.004]	theta <= 0.02
H5	Metric divergence	0.0630	[-0.022, 0.156]	theta <= 0
H6	Model-class null	-0.0165	[-0.031, -0.001]	-0.02 <= theta <= 0.02 (equivalence; interval null, two

Table 3: Supporting detail for the confirmatory family

H	note
H1	label spread 0.0123 [0.003, 0.032]; model spread 0.1499 [0.134, 0.171]
H2	6 of 22 covariates unstable (2 sign reversals, 5 magnitude >=2x). Amendment 7 diagnostic: 3 of the 6 flagg
H3	0 onsets in 179,003 person-hours; exact 95% upper bound on the per-hour onset rate = 1.67e-05 (1 in 59,75
H4	the two test sets are resampled separately and both bootstrap streams are propagated into the difference. T
H5	external AUROC 0.6960 on 29 external events (BELOW the 100-event interpretability floor); Utility Score a
H6	paired: both models scored on the same person-hours. Interval null tested at both boundaries (Amendment

Each subgroup is contrasted against the largest level of its own variable (the reference), and the resulting family of differences is corrected by Benjamini–Hochberg. The reference level is chosen by sample size, not by outcome, and is recorded in the output.

0.6 Exploratory Family E2: Subgroup Label Sensitivity

The fraction of stays labelled septic under each variant is a stay-level proportion, so the bootstrap resamples whole stays within subgroup. Because the same 65,078 stays are labelled under all three variants, the A/B/C comparison is paired within replicate, and the interval on the A–C *range* correctly reflects that pairing.

0.7 Exploratory Family E3: Alternative-Label Contrasts

Families E1 and E2 vary the label only *within* Sepsis-3, so they cannot say whether the label sensitivity is empirical or definitional. This family attaches intervals to the Protocol Amendment 2 comparison: Variant E is Sepsis-3's treatment anchor with the organ-dysfunction requirement removed, Variant D is the organ-dysfunction limb with no treatment timestamp at all. Protocol Amendment 4 adds Variant F, which keeps the Sepsis-3 conjunction intact and changes only the onset timestamp to the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$; its contrast against B is therefore a pure timing effect. Adding it enlarges family E3, so every BH-adjusted p-value in this family is computed over the larger set and is correspondingly stricter.

All variants are labelled on the same test stays, so one replicate draw applies to all of them and each contrast is paired within replicate. Reported alongside the differences is the **anchor-attributable fraction** $\phi = (\text{AUROC}_E - 0.5)/(\text{AUROC}_B - 0.5)$, the share of Variant B's above-chance discrimination that the treatment anchor reproduces on its own. ϕ is a ratio with no meaningful null at zero, so it is reported as an interval and is not part of the corrected family.

Table 4: Subgroup AUROC by race with 95% stratified cluster-bootstrap intervals (Variant B, temporal test)

model	subgroup	n_rows	n_events	auroc	CI	width
gbt	WHITE	317688	597	0.739	[0.719, 0.761]	0.042
gbt	UNKNOWN	107158	235	0.773	[0.742, 0.803]	0.061
gbt	BLACK/AFRICAN AMERICAN	31716	58	0.749	[0.676, 0.817]	0.141
gbt	OTHER	22825	35	0.669	[0.571, 0.756]	0.185
gbt	WHITE - OTHER EUROPEAN	11768	18	0.672	[0.566, 0.780]	0.214
gbt	ASIAN	6543	16	0.765	[0.625, 0.886]	0.261
gbt	HISPANIC/LATINO - PUERTO RICAN	6462	16	0.624	[0.448, 0.792]	0.343
gbt	ASIAN - CHINESE	5386	13	0.601	[0.398, 0.802]	0.404
primary	WHITE	317688	597	0.756	[0.736, 0.777]	0.041
primary	UNKNOWN	107158	235	0.773	[0.740, 0.804]	0.064
primary	BLACK/AFRICAN AMERICAN	31716	58	0.759	[0.685, 0.825]	0.140
primary	OTHER	22825	35	0.724	[0.631, 0.806]	0.175
primary	WHITE - OTHER EUROPEAN	11768	18	0.730	[0.600, 0.846]	0.246
primary	ASIAN	6543	16	0.841	[0.726, 0.929]	0.203
primary	HISPANIC/LATINO - PUERTO RICAN	6462	16	0.702	[0.546, 0.845]	0.299
primary	ASIAN - CHINESE	5386	13	0.701	[0.511, 0.869]	0.359

Table 5: Subgroup AUROC differences vs. the reference group, Benjamini-Hochberg corrected across the whole exploratory family E1

model	subgroup	reference	n_events	delta	CI	p	q_BH	significant_bh
gbt	UNKNOWN	WHITE	235	0.033	[-0.003, 0.069]	0.067	0.134	FALSE
primary	UNKNOWN	WHITE	235	0.017	[-0.022, 0.055]	0.387	0.580	FALSE

Table 6: Percentage of stays labelled septic by race and label variant, with 95% stratified cluster-bootstrap intervals

subgroup	n_stays	A	B	C	range
WHITE	40212	9.4 [9.1, 9.7]	9.6 [9.3, 9.9]	18.6 [18.3, 19.0]	9.3 [8.9, 9.7
UNKNOWN	7264	11.6 [10.9, 12.4]	12.8 [12.0, 13.6]	23.9 [22.9, 25.0]	12.3 [11.3, 1
BLACK/AFRICAN AMERICAN	4929	9.3 [8.5, 10.1]	9.1 [8.3, 9.9]	19.0 [18.0, 20.1]	9.9 [9.0, 11.
OTHER	2254	8.8 [7.7, 10.0]	9.8 [8.6, 11.0]	17.5 [15.9, 19.0]	8.7 [7.1, 10.
UNABLE TO OBTAIN	1629	9.7 [8.3, 11.2]	11.1 [9.6, 12.6]	22.5 [20.4, 24.4]	12.8 [10.7, 1
WHITE - OTHER EUROPEAN	1589	9.4 [8.1, 10.9]	7.7 [6.4, 9.0]	16.6 [14.8, 18.6]	8.9 [7.2, 10.
ASIAN	799	10.5 [8.4, 12.8]	11.5 [9.4, 13.8]	21.3 [18.4, 24.3]	10.8 [8.3, 13
ASIAN - CHINESE	721	10.1 [8.0, 12.3]	11.1 [9.0, 13.6]	19.0 [16.4, 22.1]	8.9 [6.4, 11.

Table 7: Difference in label sensitivity (A–C range) vs. the reference group, Benjamini-Hochberg corrected across exploratory family E2

subgroup	reference	n_stays	delta_pp	CI	p	q_BH	significant_b
UNKNOWN	WHITE	7264	3.01	[1.93, 4.06]	0.001	0.006	TRUE
UNABLE TO OBTAIN	WHITE	1629	3.49	[1.38, 5.55]	0.002	0.009	TRUE
BLACK/AFRICAN AMERICAN	WHITE	4929	0.62	[-0.35, 1.87]	0.186	0.347	FALSE
ASIAN	WHITE	799	1.48	[-1.04, 4.50]	0.249	0.407	FALSE
OTHER	WHITE	2254	-0.63	[-2.26, 0.98]	0.425	0.510	FALSE
WHITE - OTHER EUROPEAN	WHITE	1589	-0.41	[-2.10, 1.41]	0.635	0.714	FALSE
ASIAN - CHINESE	WHITE	721	-0.40	[-3.02, 2.61]	0.675	0.714	FALSE

Table 8: Anchor-attributable fraction: share of Variant B's above-chance AUROC reproduced by the treatment anchor alone

model	auroc_B	auroc_E	phi_anchor	CI
primary	0.7590	0.7697	1.041	[0.990, 1.097]
gbt	0.7424	0.7665	1.099	[1.037, 1.164]

Table 9: AUROC difference vs. Sepsis-3 (Variant B), Benjamini-Hochberg corrected across exploratory family E3

variant	label	model	delta	CI	p	q_BH	significant_bh
D	Deterioration	primary	0.0904	[0.073, 0.108]	0.0010	0.0020	TRUE
F	Sepsis-3 Early-Onset	primary	0.0361	[0.026, 0.046]	0.0010	0.0020	TRUE
D	Deterioration	gbt	0.1723	[0.155, 0.188]	0.0010	0.0020	TRUE
E	Anchor-Only	gbt	0.0241	[0.010, 0.038]	0.0020	0.0030	TRUE
E	Anchor-Only	primary	0.0107	[-0.003, 0.024]	0.1159	0.1391	FALSE
F	Sepsis-3 Early-Onset	gbt	0.0033	[-0.009, 0.015]	0.5587	0.5587	FALSE

Table 10: Paired ablation contrast (ablated minus full) with 95% stay-level cluster-bootstrap intervals

variant	model	full	ablated	delta	delta_ci	excludes_zero
A	gbt	0.7455	0.7418	-0.0037	[-0.0086, 0.0012]	FALSE
B	gbt	0.7424	0.7359	-0.0066	[-0.0130, -0.0001]	TRUE
C	gbt	0.7550	0.7516	-0.0034	[-0.0061, -0.0007]	TRUE
A	primary	0.7713	0.7633	-0.0080	[-0.0123, -0.0034]	TRUE
B	primary	0.7590	0.7514	-0.0076	[-0.0131, -0.0022]	TRUE
C	primary	0.7695	0.7602	-0.0094	[-0.0129, -0.0059]	TRUE

Table 11: Paired cross-label calibration contrasts against Variant B

variant	reference	delta_slope	slope_ci	slope_excludes_zero	delta_int	int_ci	int_exclu
A	B	-0.0424	[-0.114, 0.027]	FALSE	0.0286	[-0.052, 0.103]	FALSE
C	B	0.0291	[-0.031, 0.085]	FALSE	0.0710	[0.009, 0.133]	TRUE

0.8 Intervals Added Under Protocol Amendment 10

Three quantities were reported as point estimates in earlier rounds, and each carried a claim in the thesis prose that a point estimate cannot support: that dropping the action-derived features leaves discrimination "essentially unaffected", that the three label definitions make no practical difference to calibration, and that the frozen models degrade on transport to the full external cohort under Variant D. Amendment 10 attaches an interval to each. None of them enters a corrected family: the ablation arm and the Variant D arm are excluded from every family by Amendments 5 and 9 respectively, and the calibration contrasts are estimation, not testing.

0.8.1 Action-Derived Feature Ablation (Amendment 5 arm)

Both arms are fitted on the same training stays and scored on the same test stays, so the difference is paired within replicate and the interval is on the difference rather than on two marginal estimates that happen to overlap.

0.8.2 Calibration Across Label Definitions

The calibration intercept and slope are refit inside each replicate on the same stays as every other quantity here, so the cross-label differences are paired. The claim under test is the thesis's, that the label definition does not change how faithfully the fitted hazard tracks the observed rate; the contrast is therefore each variant against the primary label B.

0.8.3 Variant D Transport to the Full External Cohort (Amendment 9 arm)

The internal and external cohorts are disjoint databases, so the two bootstrap streams are drawn separately and differenced, exactly as for H5. This arm is the only external estimate in the study that clears the interpretability floor, and it was the only one carrying no interval.

```
##            used  (Mb) gc trigger   (Mb) limit (Mb)  max used   (Mb)
## Ncells  1599866  85.5    2941892  157.2         NA   2498367  133.5
## Vcells 95591571 729.4  271546524 2071.8      16384 271545773 2071.8
```

0.9 Subgroup Interval Widths

The point of attaching intervals to the equity tables is that a subgroup with few events cannot support a claim in either direction. This figure plots each subgroup AUROC with its interval against the number of sepsis onsets in that subgroup.

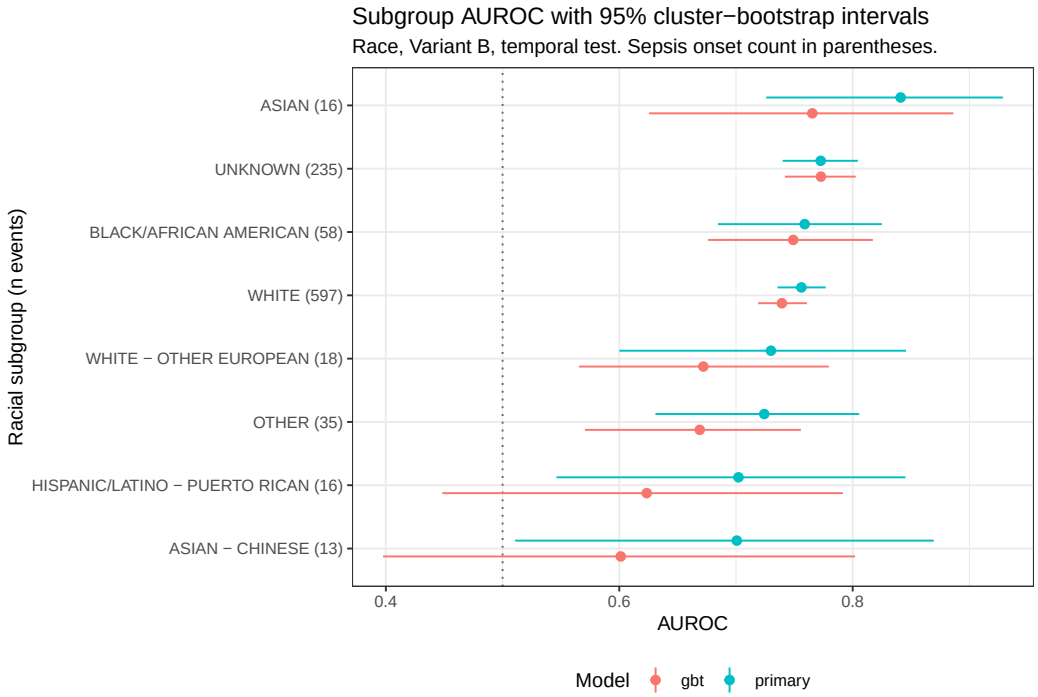


Figure 1: Subgroup AUROC with 95% cluster-bootstrap intervals (Variant B, temporal test). Interval width is driven almost entirely by the subgroup event count; the smallest groups cannot distinguish their AUROC from chance.

0.10 Analysis Register

Every analysis reported in the thesis is tagged here. `confirmatory` means the estimand, threshold, and direction were fixed in `00_preregistration.md` before any modelling. `exploratory` means the analysis was pre-registered in *scope* but not in specific contrasts, so it generates hypotheses rather than testing them. `descriptive` means no inferential claim is attached.

0.11 Reading the Corrected Family

Four things govern how the tables above should be read.

1. **The family is deliberately larger than the number of testable hypotheses.** Holm runs at $m = 6$, the count pre-registered, not the count that survived contact with the data. H2 and H3 hold slots without supplying a p-value. A hypothesis that becomes untestable after unblinding must not loosen the correction on the ones that remain, so the cost of that choice is carried here rather than passed to the surviving hypotheses.
2. **A non-confirmation is not a null result.** Failing to meet a pre-registered threshold and finding no effect are different outcomes, and the direction of a miss carries information the p-value does not. Where an estimate lands on the opposite side of its null from the one predicted, the interval is what shows it; the verdict column alone will not.

Table 12: Analysis register: confirmatory vs. exploratory status and the multiplicity correction applied to each

analysis	status	prereg
Cohort description by label variant	descriptive	<U+00A7>2
Sample size adequacy (Riley et al.)	confirmatory	<U+00A7>10
Internal temporal performance (all models)	confirmatory	<U+00A7>5, <U+
Calibration intercept and slope	confirmatory	<U+00A7>5
Decision-curve net benefit	confirmatory	<U+00A7>6
Operating points at fixed sensitivity	exploratory	not specified (thres
Per-patient detection and lead time	confirmatory	<U+00A7>6
H1 label dominance	confirmatory	<U+00A7>7
H2 instability uncertainty diagnostic	exploratory	Amendment 7 (pos
H2 coefficient instability	confirmatory	<U+00A7>7
H3 treatment leakage	confirmatory	<U+00A7>7
H4 split optimism	confirmatory	<U+00A7>7
H5 metric divergence	confirmatory	<U+00A7>7
H6 model-class null	confirmatory	<U+00A7>7
External validation on eICU-CRD	confirmatory	<U+00A7>4
Variance decomposition table	confirmatory	<U+00A7>1
Proportional-hazards (Schoenfeld) testing	confirmatory	<U+00A7>8
GBT feature importance	exploratory	not specified
Subgroup AUROC by race/language/insurance	exploratory	<U+00A7>9 (scop
Subgroup label sensitivity across variants	exploratory	<U+00A7>9 (scop
Label sensitivity vs. AUROC association	exploratory	not specified
Alternative-label AUROC vs. Sepsis-3 (variants D, E)	exploratory	Protocol Amendme
Standard onset timing min(t_susp, t_SOFA) (variant F)	exploratory	Protocol Amendme
Pre-treatment onsets under standard onset timing	descriptive	Protocol Amendme
Anchor-attributable fraction of discrimination	exploratory	Protocol Amendme
Onset agreement between label definitions	descriptive	Protocol Amendme
Subgroup interpretability floor (family E1 reduction)	exploratory	Protocol Amendme
Action-derived feature ablation (Amendment 5)	exploratory	Protocol Amendme
Variant D external validation on the full eICU-CRD cohort (Amendment 9)	descriptive	Protocol Amendme
Variant D internal-to-external transport interval (Amendment 10)	descriptive	Protocol Amendme
Ablation contrast intervals (Amendment 10)	exploratory	Protocol Amendme
Cross-label calibration contrast intervals (Amendment 10)	exploratory	Protocol Amendme

3. **Non-inferiority is read off the interval, not the test.** For H6 the question is whether the confidence limit falls inside the pre-registered margin. A large p-value on its own establishes nothing, because failing to reject is not evidence of equivalence. H6's null is an interval with two boundaries, and Amendment 10 tests both: the upper limb is large whenever the estimate is negative, so it is the lower limb that carries whatever information the test has.
4. **Exploratory families generate hypotheses and do not test them.** E1, E2 and E3 are corrected by Benjamini–Hochberg within themselves and never pooled with the confirmatory family. A contrast surviving BH is a candidate worth a future study, not a finding this design can claim.

0.12 Summary

The confirmatory family as it stands after correction, printed for the run log. The thesis reads these values from the saved tables, not from this output.

```
## Confirmatory family (Holm, m = 6, FWER = 0.05):

##   H1  Label dominance            est=-0.1375  p_raw=  1  p_holm=  1  -> NOT confirmed
##   H2  Coefficient instability    est=    6  p_raw=--  p_holm=--  -> Descriptive: instability
##   H3  Treatment leakage          est=    0  p_raw=--  p_holm=--  -> Structural: outcome abser
##   H4  Split optimism             est=-0.01654  p_raw=  1  p_holm=  1  -> NOT confirmed
##   H5  Metric divergence          est=0.06298  p_raw=0.092  p_holm=0.55  -> Not evaluable as p
##   H6  Model-class null           est=-0.01654  p_raw=0.67  p_holm=  1  -> H6 not rejected

##
## Exploratory family E1: 6 subgroup AUROC contrasts, 0 significant at BH q < 0.05
##   Smallest raw p = 0.0110 (Private, gbt vs Medicare); smallest BH q = 0.0660
##   Median 95% CI width for subgroup AUROC: 0.185 (max 0.504)

##
## Exploratory family E2: 18 label-sensitivity contrasts, 4 significant at BH q < 0.05

##
## Exploratory family E3: 6 alternative-label contrasts, 4 significant at BH q < 0.05
##   D (Deterioration) vs B, primary delta=+0.0904 [0.073, 0.108]  q_BH=0.0020
##   F (Sepsis-3 Early-Onset) vs B, primary delta=+0.0361 [0.026, 0.046]  q_BH=0.0020
##   D (Deterioration) vs B, gbt     delta=+0.1723 [0.155, 0.188]  q_BH=0.0020
##   E (Anchor-Only) vs B, gbt     delta=+0.0241 [0.010, 0.038]  q_BH=0.0030
##   E (Anchor-Only) vs B, primary delta=+0.0107 [-0.003, 0.024]  q_BH=0.1391
##   F (Sepsis-3 Early-Onset) vs B, gbt     delta=+0.0033 [-0.009, 0.015]  q_BH=0.5587

##
## Inference notebook complete.
```